Settings tested: step from rest to 460 RPM, mean of repeats

ID	What changed	Kp	Ki	Filter α	PWM jitter (% duty)	vs baseline	Overshoot (%)	Rise 10–90 % (s)	Settling to ±5 % (s)	RPM std (RPM)	RPM std vs baseline	Power (W)	runs
A	Baseline (original design)	2.629	12	1.0 (off)	**0.166**	—	**17.3**	**1.30**	4.32	**2.03**	—	4.10	5

PWM jitter = RMS sample-to-sample change of the PWM command over the last 40 % of the run. Blue row = baseline. Row colour = startup overshoot: green ≤ 5 %, amber 5–15 %, red > 15 %.